then "delivered" directly to the destination. These codes are discarded after their use and a new code is formed. Also, all parties who are to use the code must know the code and its meaning before the scheduled (or expected) broadcast of the message.

Till now we've spoken about condensed text messages, examples of codebooks and one-time codes. You still don't know what a cipher is though, and where do you draw the line between ciphers and codes?

Codes and Ciphers – the difference

Ciphers are a way to scramble messages so that other people don't understand them. They differ from codes in multiple ways. Let's list them:

1. Codes work on larger chunks, usually words. Ciphers work on lower levels. Classical ciphers work with individual letters while modern ciphers work with individual bits.
2. while ciphers are algorithms. Codes involve mapping from one language to another (encoded to decoded) and vice versa. On the other hand, ciphers are mathematical algorithms, often very complex. As such, codes would need a codebook while ciphers won't.
3. Cryptanalysis is usually not possible for wisely encoded messages. Since there's no mathematical algorithm involved, it might not be possible to find patterns in the code. For example, a code might specify that "all words starting with a vowel are to be discarded first before beginning the actual decoding process." Such a mechanism makes it nearly impossible to guess the second step involved while still containing a fully valid message (using a codebook where all encoded words begin with a consonant). Cryptanalysis depends on analyzing scrambled messages, but that's not possible with codes since they may not be broken using techniques of cryptanalysis.
4. Codes depend on the person who compiles the code and the codebook itself. The strength of the code depends on the technique he utilizes as well as his intelligence. If the code designer wasn't intelligent enough, the code's strength would be weak. On the other hand, ciphers depend on already designed mathematical procedures and their design doesn't rely, time and again, on humans.

Above all those differences is the fact that all ciphers require at least one key, while codes don't. While some may say that the encoding-decoding process also requires a codebook which is analogous, they're not similar enough to be compared. The process of coding is based on mapping– you